D02 · Supply Planning & Scheduling

02 · Constrained supply & allocation planning

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Constrained supply and allocation planning decides who gets product when demand exceeds capacity — applying explicit, governed allocation rules based on profitability, strategic priority, or contractual commitment — instead of defaulting to first-come-first-served or the loudest customer, and instead of letting the rationing rule quietly teach customers to inflate their orders.

2 WHY IT MATTERS

Cachon & Lariviere 1999 **(Cachon & Lariviere, 1999 — peer-reviewed)** showed that the allocation rule itself shapes customer behavior: when buyers know supply will be rationed proportionally to their orders, they rationally inflate orders to claim a larger share, distorting the demand signal and degrading total supply-chain performance. An allocation process that is implicit — first-come, or whoever escalates hardest — is therefore not neutral; it actively trains the channel to game it. The cost is double: product flows to the wrong customers on margin and strategic value, and the inflated orders **corrupt the very demand signal planning depends on**, amplifying the shortage in the next cycle.

3 HOW IT IS DONE

1. **Make the constraint explicit before allocating.** Quantify the true supply-demand gap by product, period, and location. Allocation only has meaning against a known shortfall; allocating against a vague constraint just spreads the guesswork.
2. **Define a governed allocation rule.** Choose and document the basis — margin, strategic-customer priority, contractual service level, fair-share — and apply it consistently. The rule must be a decision leadership owns, not an outcome that emerges from whoever escalates loudest.
3. **Allocate on true demand, not inflated orders.** Where a proportional rule would invite order inflation, allocate on a cleaned demand signal or historical sell-through instead of submitted orders, removing the incentive to game.
4. **Use available-to-promise (ATP) to enforce the allocation.** Wire the allocation into ATP/CTP so the order-promising system honors it automatically, rather than relying on planners to police every order manually under pressure.
5. **Communicate the rule to the channel.** A transparent, stable allocation policy reduces gaming because inflation no longer pays. Secrecy invites escalation; a known rule lets customers plan instead of fight.
6. **Review allocation outcomes against strategy.** Each constrained cycle, check who actually received product and whether it matched the intended priority. Allocation drift — high-margin or strategic accounts under-served while squeaky wheels are filled — is the symptom that the rule is not being enforced.

WORKED EXAMPLE — ILLUSTRATIVE

A specialty manufacturer rationed a constrained line first-come and by escalation — **baseline: orders inflated an estimated 25–40% during shortages, top-margin accounts under-served while the loudest were filled, demand signal badly distorted**. It implemented a margin-and-strategic-priority rule allocated on cleaned demand and enforced through ATP, communicated to the channel. Results: **order inflation collapsed as gaming stopped paying, fill of strategic accounts during constraint up sharply, and the demand signal stabilized enough to plan the next cycle properly**.

IN PRACTICE

A documented, governed allocation policy

Write the allocation basis, the priority tiers, and the escalation path before the next shortage, and have leadership own it. The worst time to design an allocation rule is mid-crisis, when it will be designed by whoever shouts; the policy must exist in calm.

Allocate on cleaned demand

Strip the inflation out of submitted orders by allocating on historical sell-through or a cleaned signal where possible. This both routes product more accurately and removes the incentive that produced the inflation in the first place.

Transparency with the channel

Tell customers how allocation works. A stable, known rule converts the relationship from a shouting match into a planning input — customers who know they cannot game the rule stop trying and plan around their real allocation instead.

Post-constraint allocation review

After each constrained period, review who received product versus the intended priority and quantify the margin and strategic impact. This keeps the rule honest and surfaces the drift where escalation quietly overrode the policy.

WHAT YOU WOULD USE

CONSTRAINED PLANNING / ALLOCATION & ATP

SAP IBP Response / o9 / Kinaxis — enterprise tier-1 — Constraint-based supply planning with rules-driven allocation and integrated ATP/CTP order promising. **(SAP, o9, Kinaxis — vendor)** The choice when allocation must be enforced automatically across a complex, multi-tier order book.

Blue Yonder — fulfillment & allocation — Strong allocation and order-promising capabilities integrated with demand and fulfillment. **(Blue Yonder — vendor)** A solid fit where allocation is tightly coupled to fulfillment and channel management.

Order-promising / available-to-promise specialists — Dedicated ATP/CTP and allocation engines that layer onto an existing ERP. **(specialized vendors — vendor)** The vertical option when the gap is real-time order promising rather than full supply planning.

For operations that don't need a platform — A documented allocation rule applied in a constraint-period workbook on cleaned demand, with the rule communicated to the channel. The governance and the cleaned-demand basis deliver most of the value; automation matters only once the policy itself is sound.

4 SO WHAT

> **Operations that allocate by a governed rule on true demand send scarce product where it is worth most and stop the channel from gaming; operations that ration by escalation reward the loudest customer and corrupt the demand signal they plan on next.** Scarcity is when allocation policy matters most and when most organizations have none. The order inflation that follows an implicit rule is not customer dishonesty — it is the rational response to a rule that pays it.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México